

Article Retrieval Pipeline (Task 2)

data/processed/<domain>/articles_page_granularity_<domain>.jsonl

Step 0 — Build Article Index

BM25 (rank_bm25) · TF-IDF (sklearn) · FAISS + SentenceTransformers

Step 1 — Build Query Dataset

*24 query templates per anchor · 3 context modes
3 anchor preprocessings · test-split anchors only*

Step 2 — Retrieval

BM25

TF-IDF

Dense (FAISS)
×4 models

Source article
excluded

Step 3 — Re-ranking (Zero-shot)

*Cross-encoder · top-K input: {5, 10, 20, 50}
ms-marco-MiniLM-L-6 · ms-marco-MiniLM-L-12 · deberta-v3-base*

11-dim
ablation
sweep

Step 4 — Fine-tune Reranker (optional)

*mine (query, positive, hard-negative) triples
binary cross-entropy · train split only*

Step 5 — Evaluate

Recall@1/3/5/10/20/50/100 · MRR

article_retrieval_experiments.csv

data/research/<domain>/

Index / Query Build

Sparse Retriever

Dense Retriever

Re-ranking

Evaluation

Data Store